



Modeling peak electricity demand: A semiparametric approach using weather-driven cross-temperature response functions[☆]

J. Isaac Miller^{a,*}, Kyungsik Nam^b

^a Department of Economics, University of Missouri, United States of America

^b Energy Industry Research Group, Korea Energy Economics Institute, Republic of Korea

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ABSTRACT

We propose a novel method to model daily peak electricity demand using temperature and additional hourly and daily weather covariates, such as humidity and wind speed. Rather than enter into the temperature response function additively, the additional covariates may flexibly impact the demand response to temperature. Such flexibility allows differential responses to the actual temperature based on the heat index and wind chill factor, for example. Most notably, we find that ignoring humidity substantially underestimates the effect of high temperatures, while ignoring the effect of cloud cover overestimates the effect of low temperatures. Time of day also matters: a demand response to the same temperature may be different at different times of day. Moreover, accounting for weather-related covariates improves the model's ability to explain daily peak demand.

1. Introduction

Electric utilities and system operators must be keenly aware of daily peak electricity demand. Customers and regulators favor the constant supply of electricity over the inefficiency of oversupply, hence the need for utilities to maintain operating reserves. The uncertainty driving the need for operating reserves has grown due to the expansion of intermittent renewable power generation (solar and wind), distributed generation, the shift to telecommuting precipitated by the COVID-19 pandemic, and other factors. Maintaining reserves is costly, so utilities seek to incentivize changes in consumption patterns through demand side management.

So-called demand response programs have been developed to address demand side management. Daily peaks typically occur due to coincidences of multiple factors, such as workflow patterns leading to increased commercial and industrial use during the workweek and increased residential use on the weekends (Bradford, 2018).¹ Understanding such factors allows an electric utility to identify the responsiveness

of customers during peak usage times in order to shift consumption away from these periods. To incentivize customers to make such shifts and hence reduce demand during peak periods, the utility calculates the customer baseline load, which is the hypothetical electricity demand by the customer in the absence of demand response. The utility compensates customers based on the difference between the hypothetical demand and actual load.

In this context, an array of econometric approaches have been employed to model peak electricity demand. Techniques that take into account large uncertainties, such extreme value theory, have been applied to estimate the frequency and/or magnitude of extreme peak loads (Sigauke et al., 2013; Chan and Nadarajah, 2015; Sigauke and Nemukula, 2020). In contrast, some studies adapt reduced-form time series models of demand to fit daily series of peaks. Examples include seasonal autoregressive integrated moving average models with or without conditional heteroskedasticity utilized by Sigauke and Chikobvu (2011) and Rallapalli and Ghosh (2012).²

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* Corresponding author.

E-mail addresses: millerjisaac@missouri.edu (J.I. Miller), ksnam@keei.re.kr (K. Nam).

¹ At a household level, ownership of air conditioning, swimming pools, and clothes dryers also plays a role (Fan et al., 2017).

² Forecasting electricity load and peak demand is a related and well-studied problem with its own challenges. Methods include artificial neural networks, smooth transition models, models with Markovian or non-Markovian switching, among many others. Engle et al. (1992) and Hyndman and Fan (2009) are particularly well-known references in this literature, with the former focusing on short-term forecasts and the latter focusing on long-term forecasts. See also Weron (2014) for a review of the electricity forecasting literature, and Uniejewski and Weron (2021) and Maciejowska et al. (2021) for recent papers on forecasting, with the former emphasizing forecasts using quantile regression averaging and the latter emphasizing the role of renewables in forecasting.

We model peak electricity demand, but our approach differs from those mentioned above in the sense that we first create a structural model of high-frequency electricity demand to try to take into account the factors driving the peak. We then use the results from the high-frequency model to estimate the daily peak. Of course, the most influential driver of electricity demand is temperature, and there is an extensive literature aimed at understanding and estimating the demand for electricity in response to it. A particularly influential paper by Engle et al. (1986) maps the load to temperatures by way of a nonlinear *temperature response function* (TRF) that is estimated nonparametrically. Subsequent authors (Henley and Peirson, 1997; Valor et al., 2001; Pardo et al., 2002; Moral-Carcedo and Vicéns-Otero, 2005; Chang et al., 2014); *inter alia* have improved on the basic concept of a TRF.

We utilize the concept of the *cross-temperature response function* (cross-TRF, hereafter) introduced by Chang et al. (2016b). The basic idea of a cross-TRF is that electricity demand may respond to temperature differently under different circumstances or conditionally on covariates that represent those circumstances. For Henley and Peirson (1998), Chang et al. (2016b), and subsequent authors who have employed cross-TRFs to model low-frequency electricity demand (Hong and Kim, 2018; Park and Hong, 2018), those circumstances are economic. Yet Sailor and Muñoz (1997), Yan (1998), and many others have pointed out that temperature is not the only environmental variable that affects electricity demand. In fact, at the high frequencies needed to understand peak demand, low-frequency economic variables are much less relevant than environmental variables.

Meteorologists ubiquitously remind us of the heat index and wind chill factor, making outside temperatures feel hotter and colder respectively than they really are. Buildings require more energy to stay cool at a hot temperature on a humid day than at the same temperature on a dry day. Recognizing this fact, Train et al. (1984) consider a discomfort index generated by humidity and temperature-based variables. At the other extreme of the temperature spectrum, buildings require more energy to stay warm at a cold temperature on a windy day than at the same temperature on a calm day. On a balmy late spring day, open windows invite the natural air conditioning of wind, but the howling of wind through leaky windows over a frigid night is a drain on heating.

A number of studies, including those by Li and Sailor (1995), Robinson (1997), Sailor and Muñoz (1997), Sailor (2001), Mirasgedis et al. (2006), Lam et al. (2008), Paniagua-Tineo et al. (2011), and Apadula et al. (2012), consider some combination of not only relative humidity and wind speed, but also precipitation. In addition to these three, we include cloud cover as a proxy for the amount of sunlight. Much of what drives high-frequency changes in temperature is sunlight, so once we control for temperature, it seems that the amount of sunlight may play no role. However, buildings may require more energy to stay cool on a hot sunny day than on a hot cloudy day at the same temperature due to inefficiencies in window insulation. For the same reason that house cats prefer window sills on cold sunny days, electric space heaters do not have to work as hard on such days. Moreover, natural sunlight may substitute for some electric lighting on days with few clouds.

Altogether, the environmental variables that we consider in addition to temperature, which we also refer to as weather covariates, are humidity, wind speed, cloud cover, and precipitation. We follow the literature in initially including precipitation, but we subsequently drop it based on our preliminary results. Indeed, once we control for humidity and cloud cover, it is difficult to explain how daily precipitation would play a major role in daily or hourly electricity consumption outside of some specific uses such as pumping water either out of a low-lying area (New Orleans, e.g.) when precipitation is high or into a naturally arid region (Southern California, e.g.) when precipitation is low. In these cases, precipitation would not have an interactive effect on the TRF.

We define the weather covariates relative to their typical values when temperature is near 18 °C and electricity usage is expected to be minimal. Without the interaction of temperature with humidity, the

TRF treats humidity as if it were at this typical value (65.3%) for all temperatures. We find that this omission substantially *underestimates* the effect of the highest temperatures, because they have a surprisingly *negative* correlation with humidity. Similarly, without the interaction with cloud cover, the TRF treats cloud cover as if it were at this typical value (47.38%) for all temperatures. We find that such an omission *overestimates* the effect of low temperatures, because they are often associated with clearer days, a positive correlation between temperature and cloud cover. In contrast, the omission of wind speed does not have a noticeable impact. These impacts also vary by time of day. The TRF model with no weather covariates overestimates demand response in the morning but underestimates demand response in the afternoon.

It is important to emphasize the novelty of our analysis. Our methodology consists of two distinct parts: estimating a cross-TRF for hourly demand and embedding the cross-TRF into a model of daily peak demand. Although the methodology used in the first part is adapted from that of Chang et al. (2016b), there are important differences that highlight the marginal value of our contribution. One such difference is that while those authors allow for the interaction of low-frequency economic covariates with high-frequency temperature to model low-frequency demand, we allow for the interaction of high-frequency covariates with high-frequency temperature to model high-frequency demand that may vary by time of day, a natural but non-trivial extension from a mixed-frequency to a single-frequency model. Another such difference is our introduction of environmental variables to the cross-TRF. A third difference is that the methodology used in the second part, embedding the cross-TRF into a model of daily peak demand and in the process addressing estimation bias from using the max function of an expected value, is novel to the peak demand literature to the best of our knowledge. The rest of the paper is organized as follows. We introduce the basic econometric models in the following section. In particular, we propose a high-frequency model involving cross-TRFs to model hourly demand, and we address the nonlinearity of the max function in fitting the hourly model inside of a model of daily peak demand. In Section 3, we discuss details of the procedures employed to estimate these models. Section 4 contains a discussion of the data and main empirical findings, and Section 5 concludes. Three appendices contain additional details of the data, model specification, and uncertainty estimates not elucidated elsewhere.

2. Econometric models

The first component of our modeling approach to daily peak demand is to estimate the high-frequency (hourly) link between electricity demand, temperature, and the other weather-related covariates by way of cross-TRFs. The second component relates this relationship to daily peaks in electricity demand.

2.1. High-frequency cross-TRF models

There is a long literature on modeling a so-called *temperature response function* (TRF), to use the terminology of Engle et al. (1986), a nonlinear mapping of log electricity demand from temperature. From the practical perspective of anticipating daily peak loads, load curves map electricity demand to time of day. In this analysis, we consider a mapping from both time of day (hour) and temperature to electricity demand. We denote this mapping by $g(h, r)$, where r denotes temperature and $h = 1, \dots, 24$ indexes the hour of each day. If we fix the temperature at some value r , we get a load curve $g_r(h)$ for a hypothetical day with no temperature change. If we fix the hour at some value h , we get a TRF $g_h(r)$ for that hour, which may be estimated using data observed daily at that hour. If we pool all of the hours in a day, we get a TRF $g(r)$ that reflects a daily average.

Chang et al. (2014, 2016b) model a *monthly* excess log demand response to hourly temperatures as $\int g(r) f_\tau(r) dr$ where $f_\tau(r)$ is the

distribution of hourly temperatures in a given month τ . The distribution puts a limitation on estimation, because it must be estimated for each time period (monthly, in their case) over a large number of temperature observations (hourly, in their case). Most of the data in the present analysis are observed at the same (hourly) frequency – or at least daily – but not monthly. Because this stage of our analysis concerns only high-frequency hourly data, we eliminate the integral over the distribution of high-frequency data, estimating $g(r)$ or $g_h(r)$ directly and obviating the need for an observation-intensive nonparametric density estimator.³

Without the mixed observation frequencies of monthly and hourly and ignoring the fact that some weather covariates are observed daily, a relatively simple model of log high-frequency electricity demand otherwise similar to the TRF in Eq. (3) of Chang et al. (2016b), could be written as

$$y_{t+h/24} = g(r_{t+h/24}) + \varepsilon_{t+h/24} \quad (1)$$

for day $t = 1, \dots, T$ and hour $h = 0, \dots, 23$, where $y_{t+h/24}$ is hourly excess log electricity demand. By convention, $y_{t+23/24}$ is the last hour of day t and $y_{t+24/24} = y_{t+1}$ is the first hour of day $t + 1$.

Here, “excess” means that we have conditioned out relevant and relatively predictable covariates. In our analysis, these covariates are real GDP, which controls for long-run changes in electricity demand due to income and other macroeconomic fluctuations, and indicators for irregular days, such as weekends and holidays. Macroeconomic changes tend to occur at a much lower frequency than is of interest in the present analysis, while holidays are perfectly predictable. Empirical details of the conditioning are discussed in Appendix A.

We utilize the concept of a cross-TRF $g^k(r)$, which we adapt from Eq. (8) of Chang et al. (2016b). Specifically, an extension of the high-frequency model in Eq. (1) that employs K cross-TRFs is given by

$$y_{t+h/24} = \sum_{k=0}^K w_{t+h/24}^k g^k(r_{t+h/24}) + \varepsilon_{t+h/24}, \quad (2)$$

where $w_{t+h/24}^0 = 1$ and $w_{t+h/24}^k$ for $k > 0$ are K covariates. The cross-TRF model may be further rewritten as

$$y_{t+h/24} = w'_{t+h/24} g(h, r_{t+h/24}) + \varepsilon_{t+h/24} \quad (3)$$

by redefining g as a $(K+1) \times 1$ vector of scalar functions $(g^0, g^1, \dots, g^K)'$ and defining $w_{t+h/24} = (1, w_{t+h/24}^1, \dots, w_{t+h/24}^K)'$ as a $(K+1) \times 1$ vector containing the covariates.

Prior to implementing a model selection procedure, the components of $w_{t+h/24}$ are as follows: $w_{t+h/24}^1$ denotes hourly humidity, $w_{t+h/24}^2$ denotes hourly wind speed, $w_{t+h/24}^3$ denotes hourly cloud cover with the convention that $w_{t+h/24}^3 = w_t^3$ for all h because our cloud cover data are observed daily, and $w_{t+h/24}^4$ denotes hourly precipitation with the convention that $w_{t+h/24}^4 = w_t^4$ for all h because our precipitation data are observed daily.

The cross-TRF model with covariates in Eqs. (2) and (3) reduces to the TRF model with no covariates (TRF only model) in Eq. (1) when $K = 0$. Subsequent approximations to these models, given by Eqs. (9) and (10) respectively in Section 3.1, will be labeled Model 1 and Model 0 respectively. A comparison of these models reveals inherent biases in the TRF only model (Model 0) due to omitting relevant weather covariates from the cross-TRF (Model 1).

The equivalent models in Eqs. (2) and (3) provide one core of our analysis with all hours pooled, but we estimate 24 additional models for each hour h . To this end, we may rewrite Eq. (3) as

$$y_{h,t} = w'_{h,t} g_h(r_{h,t}) + \varepsilon_{h,t}, \quad (4)$$

for $t = 1, \dots, T$ but fixing the hour h for the TRF. We use $(y_{h,t})$ to denote a daily series with h fixed, as opposed to the hourly series $(y_{t+h/24})$, and

$w_{h,t}$, $r_{h,t}$, and $\varepsilon_{h,t}$ are similarly defined. A subsequent approximation to this model, given by Eq. (8) in Section 3.1, will be labeled Model 2.

As mentioned above, we fix h because the number of hours in a day is statistically much too small to try to estimate $g(h, r)$ by applying bivariate nonparametric or semiparametric econometric methods to $r_{t+h/24}$, similarly to Chang et al. (2016a, 2019, 2021) in the related context of energy efficiency. Those authors use a local regression method with a bivariate kernel, which weights observations local to each of its arguments. Although we do not use a local regression method, in analogy to their method, we may think of ours as a local regression in which the bandwidth for the first argument is chosen so narrowly that only a single value of h is allowed. This amounts to estimating $g_h(r_{h,t})$ individually for each hour and then examining the set of 24 estimates, rather than estimating $g(h, r)$ jointly in h and r .

2.2. Daily peak demand model

The first component of our model described above is a high-frequency time series model. However, estimating daily peak demand from hourly model estimates is problematic for several reasons. First, Korea is neither Singapore nor Norilsk: as do many populated mid-latitude regions, Korea gets both hot and cold temperatures, so that peak demand does not occur at the same time of day all year long. Second, and perhaps most importantly, the max function used to identify the daily peak from hourly demand is convex, so Jensen's inequality biases daily peak estimates downward. Third, high-frequency excess demand may have serial correlation conditional on the TRF and cross-TRFs, which exacerbates the inequality and hence the bias.

It is useful to conceptualize the model in Eq. (3) as a conditional expectation function (CEF) model, but conditional on two different information sets. Let $\mathcal{G}_{t+h/24}$ denote the information set driven by past and current random variables in the series $(w_{t+h/24})$ and $(r_{t+h/24})$, and define

$$E y_{t+h/24} | \mathcal{G}_{t+h/24} = w'_{t+h/24} g(h, r_{t+h/24})$$

so that

$$\varepsilon_{t+h/24} = y_{t+h/24} - E y_{t+h/24} | \mathcal{G}_{t+h/24}$$

holds by definition. The definition allows the model in Eq. (3) to be interpreted as a CEF model as in the textbook case using cross-sectional data.

The goal is to estimate the daily peak demand conditional on hourly observations of temperatures and weather variables. Because the logarithmic transformation of demand is monotone increasing and the transformations discussed in the appendix to create excess demand are executed at the daily frequency or lower, there is no loss in generality from modeling the daily peak excess log demand denoted by $P_t = \max_{0 \leq h \leq 23} y_{t+h/24}$ for day t , so that the object to be estimated is

$$E P_t | \mathcal{G}_{t+23/24} = E \left(\max_{0 \leq h \leq 23} y_{t+h/24} \right) | \mathcal{G}_{t+23/24}, \quad (5)$$

where $t + 23/24$ denotes the last hour of day t . The max function plays the important role of allowing the hour at which the peak occurs to vary for any reason, such as seasonality or holidays.

Conditioning any function of $y_{t+h/24}$ on $\mathcal{G}_{t+23/24}$ yields the same result as conditioning on $\mathcal{G}_{t+h/24}$, because demand responds only to current temperature and weather conditions. Having observations through the end of the day cannot explain demand at a specific hour any better than having the contemporaneous and perhaps recent past observations. Hence, we may write

$$\begin{aligned} \max_{0 \leq h \leq 23} E y_{t+h/24} | \mathcal{G}_{t+h/24} &= \max_{0 \leq h \leq 23} E y_{t+h/24} | \mathcal{G}_{t+23/24} \\ &\leq E \left(\max_{0 \leq h \leq 23} y_{t+h/24} \right) | \mathcal{G}_{t+23/24} \end{aligned}$$

where the inequality follows from Jensen's inequality and the convexity of the max function. Hence, an estimate of peak demand using the

³ There are approximately 720 h in a month but only 24 h in a day.

maximum of $E[y_{t+h/24}|G_{t+h/24}]$ and specifically the model in Eq. (3) is generally downward biased.

Define the non-negative difference

$$\begin{aligned} J_t^\varepsilon &= E \left(\max_{0 \leq h \leq 23} y_{t+h/24} \right) | G_{t+23/24} - \max_{0 \leq h \leq 23} E y_{t+h/24} | G_{t+h/24} \\ &= E \left(\max_{0 \leq h \leq 23} (E y_{t+h/24} | G_{t+23/24} + \varepsilon_{t+h/24}) \right) | G_{t+23/24} \\ &\quad - \max_{0 \leq h \leq 23} E y_{t+h/24} | G_{t+23/24} \end{aligned}$$

to be the *Jensen gap*. If excess demand were completely determined by $w_{t+h/24}$ and $r_{t+h/24}, \varepsilon_{t+h/24} = 0$ would hold at all time periods. A positive variance of $\varepsilon_{t+h/24}$ pries open the Jensen gap.

An alternative approach to avoid the gap would be to aggregate all of the data to the daily level and model daily peaks over time, in the spirit of [Sigauke and Chikobvu \(2011\)](#) and [Rallapalli and Ghosh \(2012\)](#). Indeed, cloud cover and precipitation are already thus aggregated. However, if the high-frequency relationship just described holds, aggregation creates a large bias when the right hand side of Eq. (5) is not aggregated in the same way as the left-hand side. With a nonlinear aggregation scheme like the maximum and an additive error term like that in Eq. (3), aggregating all variables in the same way is impractical if not impossible.

Another alternative would be a mixed-frequency approach, such as using a MIDAS specification ([Ghysels et al., 2004](#); [Andreou et al., 2010](#); [Miller, 2014](#)) as an approximation. The hourly-to-daily frequency ratio is quite appropriate for a MIDAS specification. But mixed-frequency approaches like MIDAS rely on time-invariant weights for the high-frequency observations, which would not be appropriate for Korean peak electricity demand, which does not always occur at or near the same time. Again, Korea is neither Singapore nor Norilsk.

Up to this point, we have neither ruled out nor explicitly considered the possibility of serial correlation in the error that might result from a stationary omitted variable, though our definition of the CEF model rules out bias in estimating the high-frequency model from such an omission. Serial correlation exacerbates the Jensen gap because the variance of $\varepsilon_{t+h/24}$ increases with serial correlation. Accounting for serial correlation ameliorates the gap.

Let $F_{t+h/24}$ denote the information set driven by past and current random variables in the series $(\varepsilon_{t+h/24})$ or, equivalently, in $(y_{t+h/24} - E y_{t+h/24} | G_{t+h/24})$. The information sets $F_{t+h/24}$ and $G_{t+h/24}$ are defined so as not to overlap, and thus $E \varepsilon_{t+h/24} | F_{t+(h-1)/24} \neq 0$ may hold even though $E \varepsilon_{t+h/24} | G_{t+h/24} = 0$. In other words, serial correlation is allowed in the error term defined by the CEF model, and $(\varepsilon_{t+h/24})$ is not assumed to be a martingale difference sequence.

Rewriting $y_{t+h/24}$ as

$$E y_{t+h/24} | G_{t+h/24} + E \varepsilon_{t+h/24} | F_{t+(h-1)/24} + u_{t+h/24}$$

with $u_{t+h/24} = \varepsilon_{t+h/24} - E \varepsilon_{t+h/24} | F_{t+(h-1)/24}$ suggests replacing $\max_{0 \leq h \leq 23} E y_{t+h/24} | G_{t+h/24}$ with

$$\max_{0 \leq h \leq 23} (E y_{t+h/24} | G_{t+h/24} + E \varepsilon_{t+h/24} | F_{t+(h-1)/24})$$

in the definition of the Jensen gap J_t^ε . Doing so results in a Jensen gap of

$$\begin{aligned} J_t^u &= E \left(\max_{0 \leq h \leq 23} y_{t+h/24} \right) | G_{t+23/24} \\ &\quad - \max_{0 \leq h \leq 23} (E y_{t+h/24} | G_{t+h/24} + E \varepsilon_{t+h/24} | F_{t+(h-1)/24}) \end{aligned}$$

replacing J_t^ε .

We may expect electricity demand to respond to persistent and positively autocorrelated environmental variables, some of which may be excluded, so that $E \varepsilon_{t+h/24} | F_{t+(h-1)/24} \geq 0$ and hence

$$\max_{0 \leq h \leq 23} (E y_{t+h/24} | G_{t+h/24} + E \varepsilon_{t+h/24} | F_{t+(h-1)/24}) \geq \max_{0 \leq h \leq 23} E y_{t+h/24} | G_{t+h/24},$$

and $0 \leq J_t^u \leq J_t^\varepsilon$. As a result, accounting for serial correlation by explicitly estimating $E \varepsilon_{t+h/24} | F_{t+(h-1)/24}$ mitigates the gap.

Given estimators $\hat{E} y_{t+h/24} | G_{t+h/24}$, $\hat{E} \varepsilon_{t+h/24} | F_{t+(h-1)/24}$, and $\hat{J}^u$ of $E y_{t+h/24} | G_{t+h/24}$, $E \varepsilon_{t+h/24} | F_{t+(h-1)/24}$, and J_t^u discussed in the next section,

$$\hat{E} P_t | G_{t+23/24} = \max_{0 \leq h \leq 23} (\hat{E} y_{t+h/24} | G_{t+h/24} + \hat{E} \varepsilon_{t+h/24} | F_{t+(h-1)/24}) + \hat{J}^u \quad (6)$$

is expected to estimate $E P_t | G_{t+23/24}$ more accurately than estimation using

$$\max_{0 \leq h \leq 23} (\hat{E} y_{t+h/24} | G_{t+h/24})$$

or

$$\max_{0 \leq h \leq 23} (\hat{E} y_{t+h/24} | G_{t+h/24} + \hat{E} \varepsilon_{t+h/24} | F_{t+(h-1)/24})$$

alone. Keeping in mind that y denotes excess log demand, the adjustment is actually a product of $\exp(\hat{J}_t^u)$ and the exponent of the estimated maximum.

The discussion of this section generalizes to an alternative estimator

$$\hat{E} P_t | G_t = \max_{0 \leq h \leq 23} (\hat{E} y_{h,t} | G_{h,t} + \hat{E} \varepsilon_{h,t} | F_{h,t-1}) + \hat{J}^u \quad (7)$$

that utilizes h models specific to each hour in a day and estimators $\hat{E} y_{h,t} | G_{h,t}$ and $\hat{E} \varepsilon_{h,t} | F_{h,t-1}$ of $E y_{h,t} | G_{h,t}$ and $E \varepsilon_{h,t} | F_{h,t-1}$ and is based on the model in Eq. (4) rather than that in Eq. (3).

3. Estimation procedures

3.1. Procedure for high-frequency cross-TRF models

3.1.1. Procedure for a specific hour

For a specific hour h , we estimate the TRF and cross-TRFs in the model in Eq. (4) using a technology from the extant literature. Specifically, letting $s = (r + 15)/55$ because all temperatures in the sample are between -15°C and $-15 + 55 = 40^\circ\text{C}$, we approximate $g_h^k(r)$ by

$$g_h^k(r) \simeq \tilde{g}_h^k(s) = \sum_{i=0}^{p_{hk}} \alpha_{h,i}^k s^i + \sum_{j=1}^{q_{hk}} [\alpha_{h,1j}^k \cos(2\pi j s) + \alpha_{h,2j}^k \sin(2\pi j s)],$$

using the well-known flexible Fourier functional (FFF) form introduced to econometrics by [Gallant \(1981\)](#) – and see also [Park and Hahn \(1999\)](#) – and used extensively in the energy economics literature ([Chang et al., 2014, 2016b](#); [Miller and Nam, 2020](#)).

The elegance and power of the FFF form is that $\tilde{g}_h$ admits a linearization in the parameters α_i , α_{1j} , and α_{2j} so that it can be estimated using linear methods such as ordinary least squares. Specifically,

$$\tilde{g}_h^k(s)w = \sum_{i=0}^{p_{hk}} \alpha_{h,i}^k [s^i w] + \sum_{j=1}^{q_{hk}} \alpha_{h,1j}^k [\cos(2\pi j s)w] + \sum_{j=1}^{q_{hk}} \alpha_{h,2j}^k [\sin(2\pi j s)w]$$

or, for a given temperature and covariate on day t at hour h ,

$$\tilde{g}_h^k(s_{h,t})w_{h,t}^k = x_{h,t}^{k'} \alpha_h^k,$$

where

$$x_{h,t}^k = [1, s_{h,t}, \dots, s_{h,t}^{p_{hk}}, \cos(2\pi s_{h,t}), \sin(2\pi s_{h,t}), \dots, \cos(2\pi q_{hk} s_{h,t}), \sin(2\pi q_{hk} s_{h,t})] \times w_{h,t}^k$$

$$\text{and } \alpha_h^k = (\alpha_{h,0}^k, \dots, \alpha_{h,p_{hk}}^k, \alpha_{h,11}^k, \alpha_{h,21}^k, \dots, \alpha_{h,1q_{hk}}^k, \alpha_{h,2q_{hk}}^k)'$$

By stacking across $k = 0, \dots, K$ and defining $\alpha_h = (\alpha_h^0, \dots, \alpha_h^{K'})'$ and $x_{h,t} = (x_{h,t}^0, \dots, x_{h,t}^{K'})'$, we may approximate the model in Eq. (4) by

$$\text{Model 2 : } y_{h,t} = x_{h,t}' \alpha_h + \eta_{h,t}, \quad (8)$$

where $\eta_{h,t}$ contains both the original idiosyncratic error $\varepsilon_{h,t}$ and any approximation error from selection of $p_{h0}, \dots, p_{hK}$ and $q_{h0}, \dots, q_{hK}$. We

assume that $\alpha_1^k = \dots = \alpha_{24}^k$ for $k > 0$ so that the TRF but not the cross-TRFs may vary by time of day. Our rationale for this set of restrictions is that once we control for time of day by way of the TRF, allowing the cross-TRFs to vary over time becomes redundant and sacrifices parameter parsimony.

Given estimates $\hat{\alpha}_h^k$ of the subvector α_h^k of α_h , the cross-TRF given by $g_h^k(r)$ for covariate k is estimated by

$$\hat{g}_h^k(r) = \sum_{i=0}^{p_{hk}} \hat{\alpha}_{h,i}^k \left(\frac{r+15}{55} \right)^i + \sum_{j=1}^{q_{hk}} \left[\hat{\alpha}_{h,1,j}^k \cos \left(\frac{2\pi j(r+15)}{55} \right) + \hat{\alpha}_{h,2,j}^k \sin \left(\frac{2\pi j(r+15)}{55} \right) \right]$$

which may be plotted and interpreted as a three-dimensional surface over h and r . In the notation of Section 2, we set $\hat{\mathbf{E}}_{y_{h,t}|\mathcal{G}_{h,t}} = w'_{h,t} \hat{g}_h(r_{h,t}) = x'_{h,t} \hat{\alpha}_h$.

3.1.2. Procedure for pooled hours

Suppose instead that we pool the model across hours. Using the same arguments as above, we approximate the model in Eq. (3) using

$$\text{Model 1: } y_{t+h/24} = x'_{t+h/24} \alpha + \eta_{t+h/24}, \quad (9)$$

where $\alpha = (\alpha^0, \dots, \alpha^{K'})'$ with $\alpha^k = (\alpha_0^k, \dots, \alpha_{p_k}^k, \alpha_{11}^k, \alpha_{21}^k, \dots, \alpha_{1q_k}^k, \alpha_{2q_k}^k)'$, where $x_{t+h/24} = (x_{t+h/24}^0, \dots, x_{t+h/24}^{K'})'$ with $x_{t+h/24}^k$ defined analogously to $x_{h,t}^k$ except using hourly $(s_{t+h/24})$ and $(u_{t+h/24}^k)$ in place of daily $(s_{h,t})$ and $(u_{h,t}^k)$, and where $\eta_{t+h/24}$ is analogous to $\eta_{h,t}$. Given estimates $\hat{\alpha}$ of α , we set $\hat{\mathbf{E}}_{y_{t+h/24}|\mathcal{G}_{t+h/24}} = x'_{t+h/24} \hat{\alpha}$ in the notation of Section 2.

When $K = 0$, the cross-TRF model in Eq. (9) simplifies to a simple TRF model given by

$$\text{Model 0: } y_{t+h/24} = x_{t+h/24}^0 \alpha^0 + \eta_{t+h/24}^0, \quad (10)$$

with $x_{t+h/24}^0$ and α^0 defined as above and $\eta_{t+h/24}^0$ defined analogously with additional approximation error. We label this last and simplest TRF only model in Eq. (10) as Model 0. We expect the richer cross-TRF specifications of Models 1 and 2 to dominate this benchmark specification.

3.2. Procedure for the daily peak demand model

Up to this point, we have estimators given by $\hat{\mathbf{E}}_{y_{t+h/24}|\mathcal{G}_{t+h/24}} = x'_{t+h/24} \hat{\alpha}$ for Model 1 (pooled hours) and $\hat{\mathbf{E}}_{y_{h,t}|\mathcal{G}_{h,t}} = x'_{h,t} \hat{\alpha}_h$ for Model 2 (specific hours). We need $\hat{\mathbf{E}}_{\varepsilon_{t+h/24}|\mathcal{F}_{t+(h-1)/24}}$ or $\hat{\mathbf{E}}_{\varepsilon_{h,t}|\mathcal{F}_{h,t-1}}$ from Eqs. (6) and (7) respectively to mitigate the Jensen gap. We estimate $\hat{\mathbf{E}}_{\varepsilon_{t+h/24}|\mathcal{F}_{t+(h-1)/24}}$ from Model 1 using

$$\hat{\mathbf{E}}_{\varepsilon_{t+h/24}|\mathcal{F}_{t+(h-1)/24}} = \hat{\rho}_1 \hat{\eta}_{t+(h-1)/24} + \dots + \hat{\rho}_d \hat{\eta}_{t+(h-d)/24},$$

where the ρ 's are estimated as in the bootstrap procedure described in Appendix C. The notation differs for Model 2 with a fixed hour, and the estimator is given by

$$\hat{\mathbf{E}}_{\varepsilon_{h,t}|\mathcal{F}_{h,t-1}} = \hat{\rho}_1 \hat{\eta}_{h,t-1} + \dots + \hat{\rho}_d \hat{\eta}_{h,t-d},$$

but the procedure is the same. The notation d denotes a generic lag order. The increments are hourly in Model 0 and Model 1, and setting $d = 48$ is sufficient to eliminate evidence of further serial correlation. However, the increments are daily in Model 2, and setting $d = 3$ is sufficient.

To estimate J_t^u , we use the moment estimator given by

$$\hat{J}_t^u = T^{-1} \sum_{i=1}^T \left[P_i - \max_{0 \leq h \leq 23} \left(\hat{\mathbf{E}}_{y_{t+h/24}|\mathcal{G}_{t+h/24}} + \hat{\mathbf{E}}_{\varepsilon_{t+h/24}|\mathcal{F}_{t+(h-1)/24}} \right) \right],$$

which approximates J_t^u as invariant over time.

We then estimate the log peak excess electricity demand on day t as

$$\hat{P}_t = \max_{0 \leq h \leq 23} \left(\hat{\mathbf{E}}_{y_{t+h/24}|\mathcal{G}_{t+h/24}} + \hat{\mathbf{E}}_{\varepsilon_{t+h/24}|\mathcal{F}_{t+(h-1)/24}} \right) + \hat{J}_t^u,$$

Table 1

Summary Statistics. Summary statistics of temperature and the weather variables before standardization.

Variable	Units	Mean	Std. Dev.	Min	Max
Temperature	°C	14.1	10.2	-14.1	37.1
Humidity	%	63.5	18.2	11.6	98.8
Wind speed	m/s	2.2	1.0	0.0	9.6
Cloud cover	%	48.3	31.3	0.0	100.0
Precipitation	mm	3.6	14.4	0.0	301.5

the linear procedures described in the appendix to create log excess demand are reversed to estimate the log of actual demand, and finally the result is exponentiated to obtain an estimate of peak demand in megawatts.

4. Data and empirical results

Data and sources

We move now to a brief discussion of the data and sources before proceeding to discuss empirical results. Most the data employed in the analysis were generously provided by the Korea Energy Economics Institute (KEEI).

We use hourly total power generation data in megawatts (MW) including the amount of demand response from January 1, 2010 through December 31, 2019 from Korea Power Exchange (KPX). Note that total power generation is the sum of auxiliary load, transmission and distribution losses, and power sold. Because auxiliary load and transmission and distribution losses are unavailable at the same frequency, we equate demand with total power generation as in many macro-level studies of electricity demand.

For a handful of countries, like Uruguay, that export a large percentage of their generation, such an assumption would be misleading. The geopolitics of the Republic of (South) Korea are such that large-scale imports or exports are impossible. Moreover, storage is negligible due to physical constraints. Henceforth, we refer to generation and demand interchangeably. The appendix contains additional details on the construction of excess energy demand.

All of the weather variables used for this analysis were obtained from KMA National Climate Data Center.⁴ They are as follows: hourly temperature, hourly humidity, hourly wind speed, daily cloud cover, and daily precipitation. The data represent five geographically distributed Korean cities: Seoul, Busan, Daejeon, Kwangju, and Daegu, and we calculate a national average by way of a simple average across the five cities. The daily data are extrapolated to the hourly frequency by assuming constancy over the 24 h of each day.

For most of the analysis, excluding the discussions of annual data following Fig. A.1 and Table A.3 in Appendix A, we standardize the weather data other than temperature relative to 18 °C, the temperature at which electricity demand is expected to be at its minimum. Table 1 shows summary statistics for the data prior to this standardization.

To implement the standardization, we first calculate an average of each of the four series in a neighborhood of 18 °C. These local averages are calculated using simple Nadaraya-Watson local regressions with a Bartlett-Epanechnikov kernel and a Silverman bandwidth (c.f. Pagan and Ullah, 1999, or Li and Racine, 2007, e.g.). Specifically, these local averages are 65.3% humidity, 2.14 m/s wind speed, 47.38% cloud cover, and 2.7 mm precipitation. These may be thought of as typical conditions over the period of an hour in which the average temperature is 18 °C, but these are not the averages of the weather variables themselves shown in the table.

⁴ Downloaded from <https://data.kma.go.kr/cmmn/main.do> on October 16, 2020.

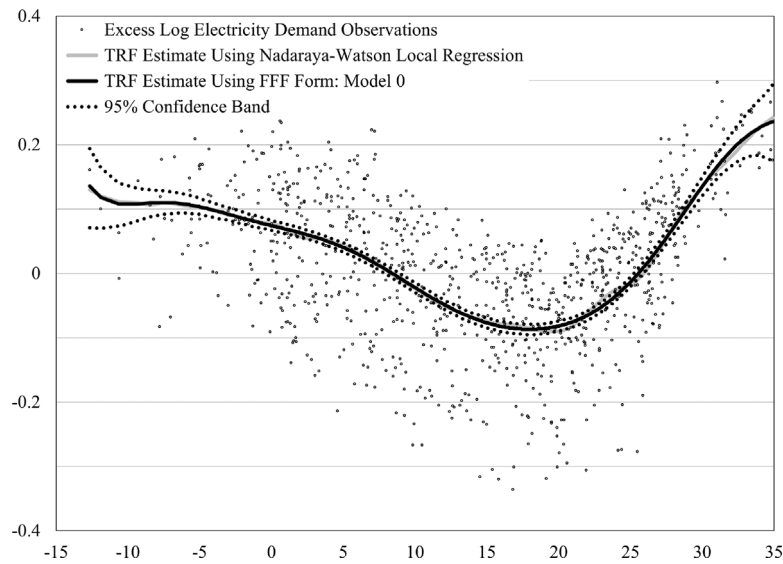


Fig. 1. Model 0: Daily TRF from the TRF Only Model. Daily pooled TRF estimated using Eq. (9) with no covariates ($K = 0$). Approximately 1% of the data are shown with TRFs estimated using the full sample with FFF form (preferred approach) compared to a Nadaraya–Watson regression for visual comparison.

With these averages in hand, we calculate the hourly deviations from these averages over the standard deviation for each of the four series. The resulting series show units of standard deviation from a local-to-18 °C norm. The TRF may be interpreted as if only temperature varies while holding variations in the other weather variables constant. If humidity increases from zero to one unit while temperature stays the same, the cross-TRF for humidity gives the excess demand response at each temperature from a standard deviation above the normed humidity of 65.3%. The standard deviations are 18.2% humidity, 0.99 m/s wind speed, 31.34% cloud cover, and 14.4 mm precipitation.

4.1. Results for the high-frequency TRF and cross-TRF models

We first estimate Model 0 in Eq. (10), the daily TRF with no cross-TRFs, as a benchmark. The first column of Table 2 shows the parameter estimates, with the Hannan–Quinn Information Criterion (HQIC) selecting $(p_{*0}, q_{*0}) = (4, 3)$ from a maximum of $(5, 3)$.

Fig. 1 shows the results in comparison with a Nadaraya–Watson local regression using the full sample against a randomly drawn fraction of the data. The Nadaraya–Watson regression shows the most flexible possible specification, but such estimators are highly inefficient in a statistical sense. The semiparametric FFF form with low-order expansion of Model 0 is a much more efficient estimator. As the figure illustrates, both techniques estimate nearly identical TRFs, suggesting that the FFF form is sufficiently flexible to capture the response of excess log electricity demand to temperature. Appendix C describes the bootstrap procedure we use to estimate the 95% confidence band in Fig. 1 and subsequent figures.

We see the expected shape of the TRF in Fig. 1. Like those of St. Louis, Georgia, or the Northeastern US in Engle et al. (1986), the TRF reaches its minimum at about 18 °C and increases steeply with air conditioning usage as the temperature rises and less steeply as temperature falls, because natural gas is a viable alternative means for heating. Such a pattern is clear from the data plotted, but they also show a sizable amount of variation not accounted for by the TRF (R^2 and $\bar{R}^2$ are 0.363).

We next estimate the daily TRF with cross-TRFs, given by Model 1 in Eq. (8) pooled across all h . For the sake of further model specification, we estimate four models with each of the covariates, humidity, wind speed, cloud cover, and precipitation, *dropped*. The respective $\bar{R}^2$ statistics are 0.407, 0.450, 0.443, and 0.458, compared to 0.459 with all five covariates. Because there is almost no improvement from

Table 2

TRF Parameter Estimates. Models 0 and 1, given respectively by Eqs. (10) and (9), use pooled hourly data to estimate daily TRF and cross-TRFs. Model 2, given by Eq. (8), uses daily data for each of 24 h. See Table 4 for lag orders selected for Model 2.

	Model 0	Model 1	Model 2			
	Daily	Daily	06:00	12:00	18:00	24:00
α_{*0}^0	-7.662	0.095	0.003	0.123	-0.125	0.177
α_{*1}^0	-8.312	-1.059	0.906	-0.632	0.494	0.125
α_{*2}^0	273.236	9.524	-4.600	10.789		-1.138
α_{*3}^0	-524.427	-32.136	4.167	-41.706		-1.635
α_{*4}^0	259.574	40.371		56.084		3.323
α_{*5}^0		-16.261		-24.444		
α_{*11}^0	7.549				0.096	
α_{*21}^0	0.295				0.171	
α_{*12}^0	0.372					
α_{*22}^0	-0.016					
α_{*13}^0	0.038					
α_{*23}^0	-0.002					

precipitation, we subsequently *drop* precipitation, so that $K = 3$ and we focus only on humidity, wind speed, and cloud cover. The parameter estimates are reported in the respective columns of Table 2 (TRF) and Table 3 (cross-TRFs), with HQIC selecting $(p_{*0}, q_{*0}) = (5, 0)$ from a maximum of $(5, 3)$. As discussed in Appendix B, we set $(p_{*k}, q_{*k}) = (1, 2)$ for $k > 0$ to avoid parameter proliferation.

Fig. 2 shows the estimated TRF for Model 1 compared to that of Model 0, in addition to the estimated cross-TRFs. Because the covariates have been standardized to their average values near 18 °C, each cross-TRF shows the additional response of electricity demand to temperature at a single standard deviation above the covariate's average value when temperature is 18 °C. The positive effect of humidity at warmer temperatures is the most salient feature, but it turns out that humidity below about 25 °C has a small but significant negative effect on demand. The effect of wind speed is not very large at any temperature, but it is significantly positive up to about 30 °C.

Finally, the effect of cloud cover is small but significantly positive between about 0 °C and about 25 °C, clearly showing the demand for lighting under cloudy conditions. The effect is insignificant at temperatures exceeding 25 °C, but significantly negative – albeit still relatively small – at the coldest temperatures. The coldest temperatures tend to be clear – i.e., negative standardized cloud cover – so demand

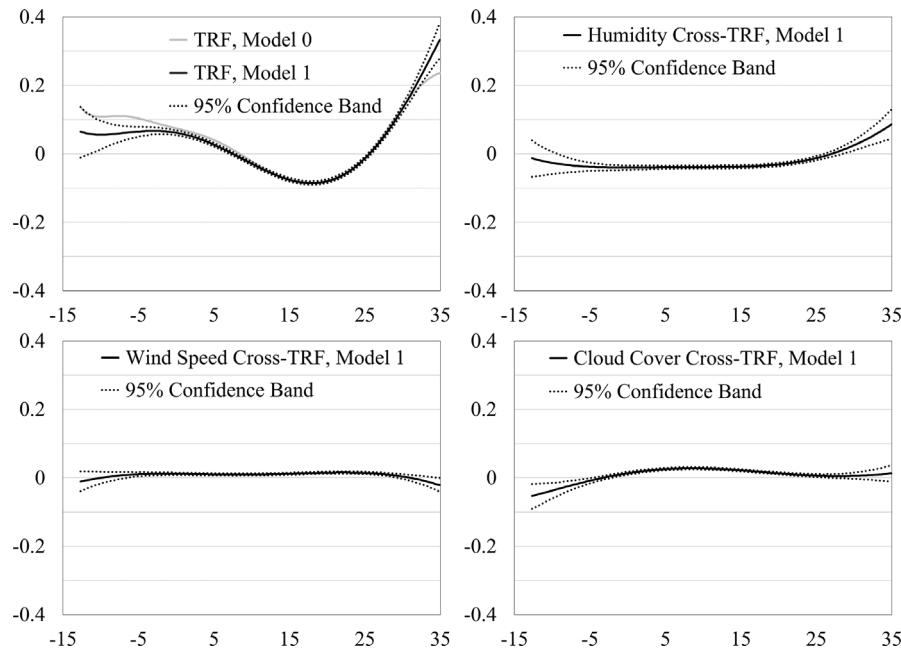


Fig. 2. Model 1: Daily TRF and Cross-TRFs. Daily pooled TRF estimated using Eq. (9) with $K = 3$ covariates.

for electricity for lighting decreases, while at the same time more distributed solar generation is available for peak heating demand.

The omission of the weather covariates (Model 0) exaggerates the effect of cold temperatures while it downplays the effect of hot temperatures. To understand why, we can fix temperature at a specific value, say -5°C or 35°C , so that Models 0 and 1 simply become linear regressions of excess electricity demand on a constant and the covariates. From textbook omitted variable bias results, the sign of the omitted variable bias is given by the product of the signs of the effect of the covariate on demand at that temperature and the mean of the covariate at that temperature.

At 35°C , Fig. 2 and solid intuition suggest that humidity increases demand, but it is less clear whether hot days tend to be more or less humid. This question is answered by Fig. 3, which shows a plot of humidity data against temperature. There is clearly a positive correlation (see also Table A.3 in Appendix A), but that correlation switches to negative above approximately 25°C and is estimated to be about a standard deviation lower on average at 35°C than at 18°C . Therefore, the hottest hours of the sample are surprisingly less humid on average than more temperate hours. We can therefore expect a negative omitted variable bias at the hottest temperatures, which is in fact what we see in the TRF of Model 0 compared to that of Model 1 in Fig. 2.

At -5°C , Fig. 2 suggests that cloud cover decreases demand. At the same time, cloud cover is estimated to be about a standard deviation lower on average at -5°C than at 18°C according to Fig. 3. The coldest hours of the sample are clearer than more temperate hours. The product of the negative signs suggests that excess electricity demand is overestimated by Model 0 at colder temperatures, which is indeed borne out in the TRFs shown of Models 0 and 1 in Fig. 2.

Finally, we estimate the hourly TRF with hourly cross-TRFs, Model 2, given by Eq. (8) but fixing h for 24 separate hourly models. Table 4 shows the lag order selected for each hour. We present further results only for the 6th, 12th, 18th, and 24th hours of each day. The right four columns of Tables 2 and 3 show the parameter estimates for the TRF and cross-TRFs respectively at these hours. The hourly TRFs and cross-TRFs for these four hours are plotted in Figs. 4–7. Note that unlike Fig. 2, which shows the daily TRF and all three cross-TRFs, Figs. 4–7 each show four hours of the TRF or four hours of each of the cross-TRFs, facilitating hourly comparisons.

Table 3

Cross-TRF Parameter Estimates. Model 1, given by Eq. (9), uses pooled hourly data to estimate daily TRF and cross-TRFs. Model 2, given by Eq. (8), uses daily data for each of 24 h.

	Model 1 Daily	Model 2 06:00	Model 2 12:00	Model 2 18:00	Model 2 24:00
$\alpha_{\cdot,0}^1$	0.034	-0.018	-0.552	-0.428	-0.034
$\alpha_{\cdot,1}^1$	-0.588	0.147	2.647	2.083	0.236
$\alpha_{\cdot,2}^1$	0.760	-0.156	-2.386	-1.879	-0.247
$\alpha_{\cdot,11}^1$	-0.032	0.005	0.173	0.140	0.018
$\alpha_{\cdot,21}^1$	0.026	-0.003	0.043	0.028	-0.009
$\alpha_{\cdot,0}^2$	-0.056	0.024	-0.111	-0.087	0.007
$\alpha_{\cdot,1}^2$	0.436	-0.133	0.551	0.486	0.014
$\alpha_{\cdot,2}^2$	-0.477	0.117	-0.527	-0.505	-0.071
$\alpha_{\cdot,11}^2$	0.032	-0.019	0.037	0.045	-0.005
$\alpha_{\cdot,21}^2$	-0.011	0.002	0.004	-0.007	-0.015
$\alpha_{\cdot,0}^3$	-0.053	-0.070	0.163	0.093	0.004
$\alpha_{\cdot,1}^3$	0.162	0.413	-0.745	-0.510	-0.069
$\alpha_{\cdot,2}^3$	-0.068	-0.435	0.664	0.497	0.106
$\alpha_{\cdot,11}^3$	-0.014	0.021	-0.050	-0.042	-0.014
$\alpha_{\cdot,21}^3$	0.025	0.004	-0.020	-0.001	0.016

Table 4

Model 2: Optimal Lag Selection. Lag orders chosen for Model 2 over each of 24 h by HQIC with a maximum of $(p_{h0}, q_{h0}) = (5, 3)$.

0h:00	p_{h0}	q_{h0}	0h:00	p_{h0}	q_{h0}	0h:00	p_{h0}	q_{h0}
01:00	3	0	09:00	1	1	17:00	1	1
02:00	3	0	10:00	1	2	18:00	1	1
03:00	3	0	11:00	1	2	19:00	3	0
04:00	3	0	12:00	5	0	20:00	3	0
05:00	3	0	13:00	3	1	21:00	2	1
06:00	3	0	14:00	3	1	22:00	2	1
07:00	3	0	15:00	2	1	23:00	4	0
08:00	3	3	16:00	2	1	24:00	4	0

Fig. 4 specifically shows the TRFs for Model 2 with that of Model 0 plotted as a benchmark. Strangely, perhaps, the hourly TRF that best resembles the daily TRF with no covariates is at midnight. In fact, this is borne out by Figs. 5–7, which show that none of the covariates

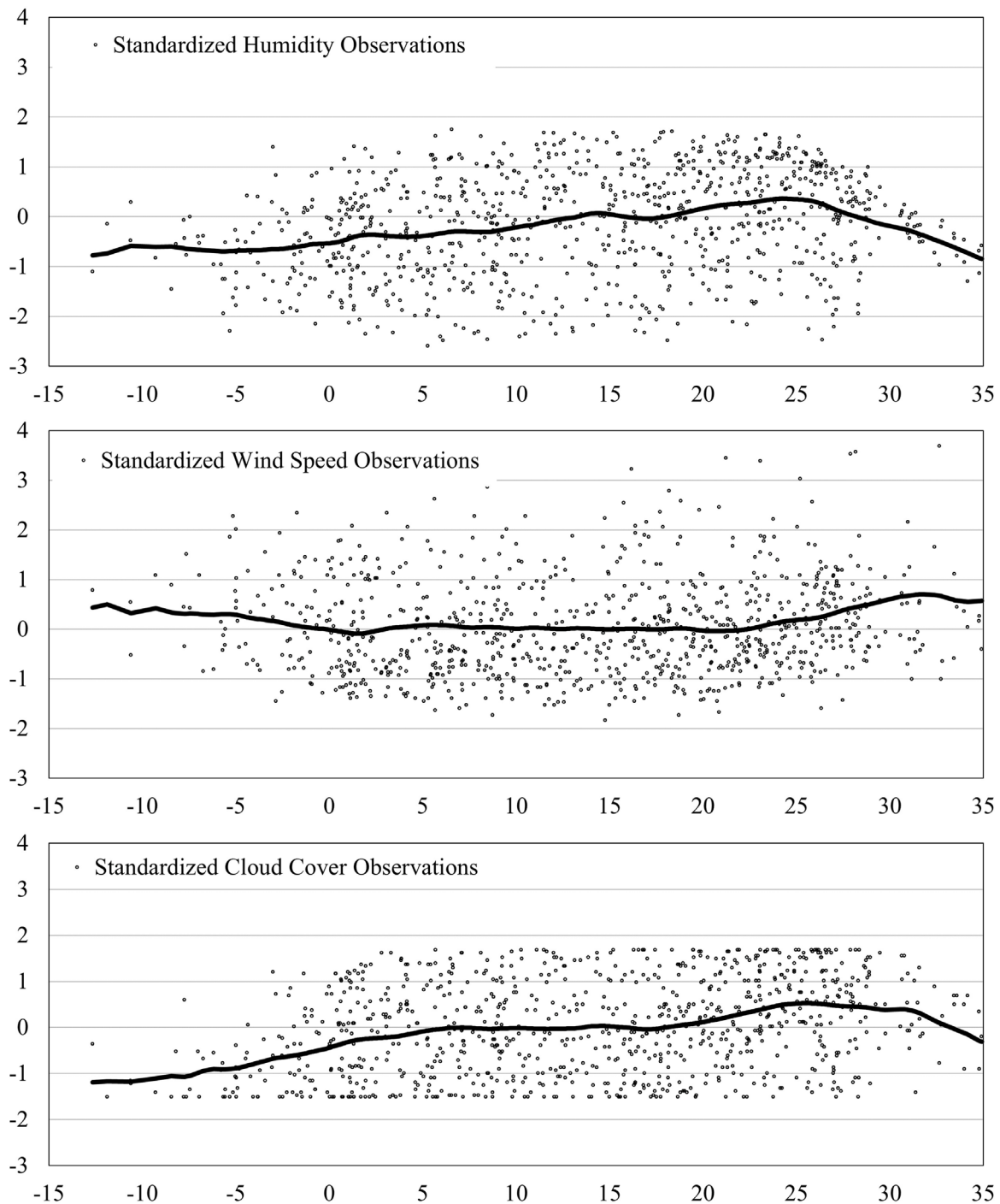


Fig. 3. Covariates vs. Temperatures. Approximately 1% of the data are shown compared to Nadaraya–Watson regressions of each covariate on temperature (solid lines) over the full sample for visual display only.

significantly influence the demand response at midnight: the cross-TRFs at midnight are essentially flat. The daily TRF in Model 0 generally overestimates demand in the morning, but generally underestimates demand at noon and in the afternoon. Because the daily peak often occurs in the late afternoon, the underestimation may substantially impact estimation of the peak. Otherwise, the shapes are quite similar across times of day.

As already mentioned, humidity does not appear to affect the demand response at midnight in Fig. 5, and this also appears to be the case in the morning. At noon and in the afternoon, the most salient effect of humidity occurs at cold temperatures. More humidity at cold temperatures dampens the effect of those temperatures on demand. Or,

the converse, more energy is demanded on arid cold afternoons than on moist afternoons of the same temperature.

The vertical axes of the plots of the wind speed cross-TRF, Fig. 6, are fixed to be the same as those of the humidity cross-TRF, so it is clear that the magnitude of the effect of wind speed is smaller. There is some indication that windy noons and afternoons have ameliorative effects on demand at hot temperatures. Indeed, wind may offset air conditioning needs, but the effect is not strongly significant.

The plots in Fig. 7 for cloud cover show nearly the opposite effects as those of wind speed. Again, neither effects at midnight nor those in the morning are apparent, but now positive effects at noon and in the afternoon are so. The positive effect seems logical: a cloudy day

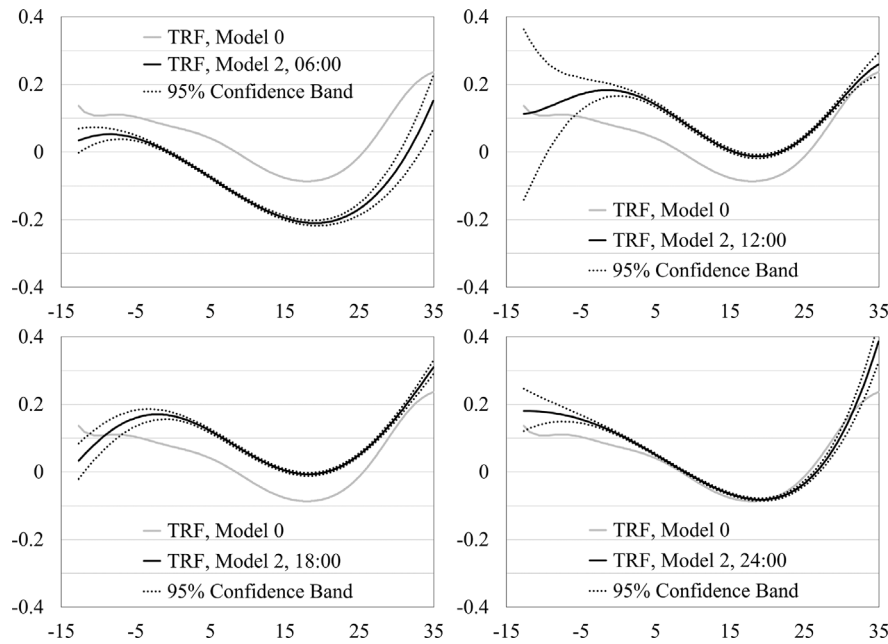


Fig. 4. Model 2: Hourly TRFs. TRFs for 06:00, 12:00, 18:00, and 24:00 estimated using Eq. (8) with $K = 3$ covariates.

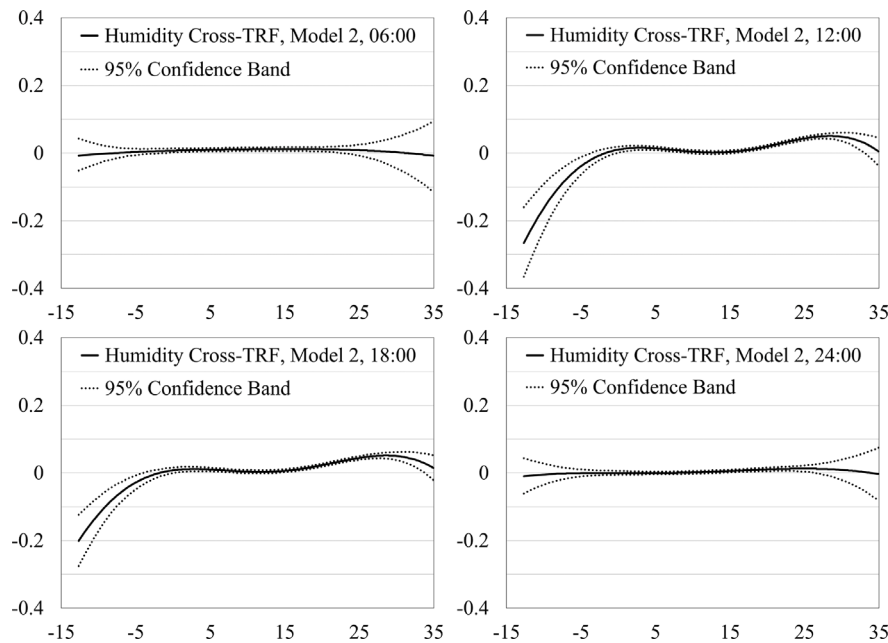


Fig. 5. Model 2: Hourly Humidity Cross-TRFs. Cross-TRFs for 06:00, 12:00, 18:00, and 24:00 estimated using Eq. (8) with $K = 3$ covariates.

increases demand for heating and lighting at the same temperature. However, these results should be taken with a grain of salt, because cloud cover is observed as a daily average rather than hourly and the estimated impacts are insignificant or barely significant.

4.2. Results for the daily peak demand model

In order to interpret the results for daily peak demand, first consider a single day: Wednesday, December 25, 2019. The actual peak that day was 65 610 MW (at 19:00). The predicted peak for that year based on GDP alone was 63 669 MW (Table A.2 of Appendix A), but the fact that it was Christmas predicts a peak of $63\,669 \times \exp(0.050 - 0.106) \approx 60\,206$ meaning there was a deficit of $65\,610 - 60\,206 = 5404$ MW. A deficit is expected because December is not a temperate month, and because we

are not yet taking into account any temperature response. In general, the models yield estimates given by $60\,206 \exp(\hat{P}_t + \hat{J}^u)$. Table 5 shows demand deficits in MW over the last week of 2019 for no model, for Models 0, 1, and 2 with neither serial correlation adjustment nor Jensen adjustment, for Model 2 with serial correlation adjustment but no Jensen adjustment (labeled Model 2+), and for Model 2 with both serial correlation and Jensen adjustments (labeled Model 2++).

It turns out that Model 0, which takes into account temperature but not the weather covariates, has the smallest deficit or surplus on December 25, 2019. However, looking at the remaining days of the week, Model 0 does not do as well as Model 1 on the other 6 days. Model 2 does better on some days but not others, and the adjustments do not make uniform improvements.

Of course, looking at just one day or week is hardly a fair comparison. Calculating the root mean squared error (RMSE) over the whole

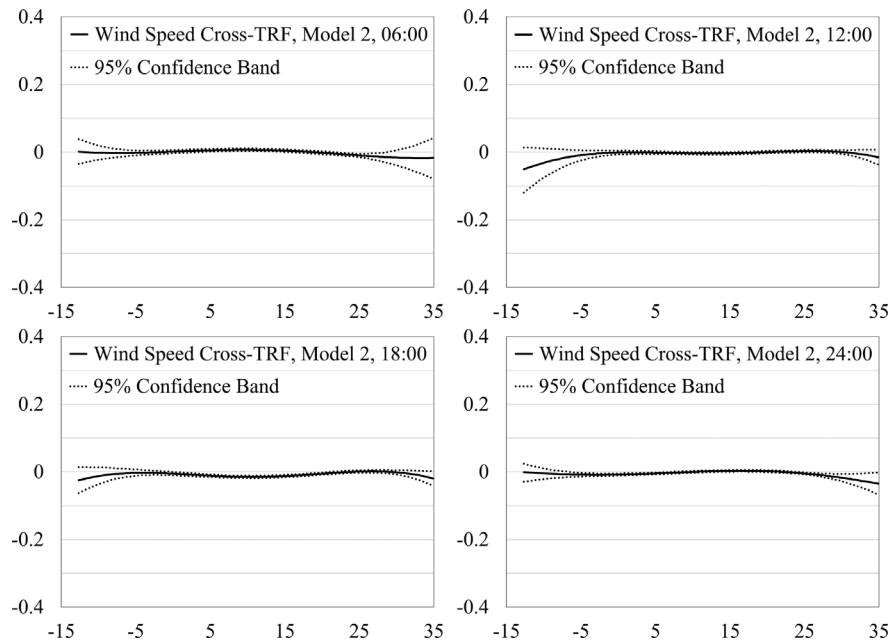


Fig. 6. Model 2: Hourly Wind Speed Cross-TRFs. Cross-TRFs for 06:00, 12:00, 18:00, and 24:00 estimated using Eq. (8) with $K = 3$ covariates.

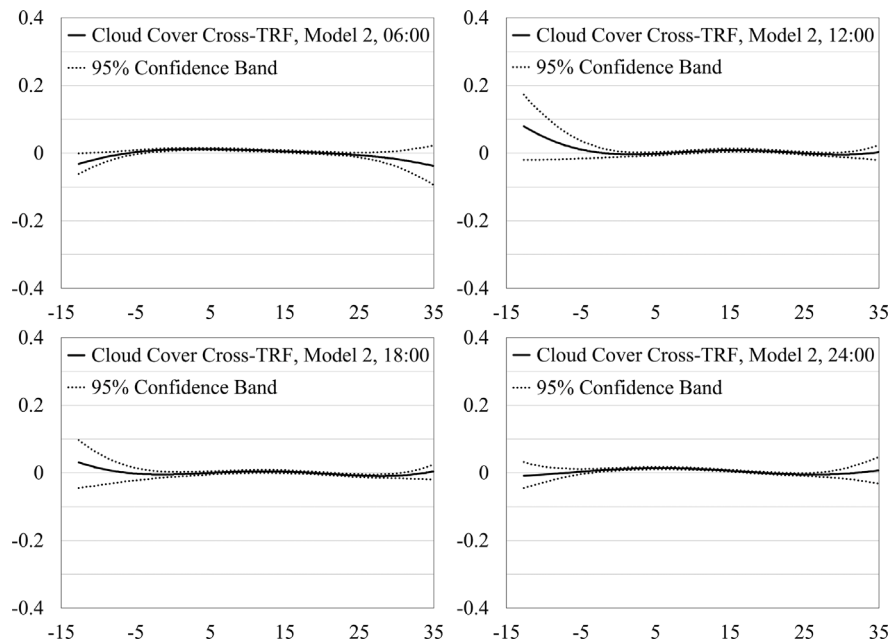


Fig. 7. Model 2: Hourly Cloud Cover Cross-TRFs. Cross-TRFs for 06:00, 12:00, 18:00, and 24:00 estimated using Eq. (8) with $K = 3$ covariates.

sample gives a clearer picture. With no model and simple demeaning, the RMSE is 8216 MW, but the TRF only model given by Model 0 improves the RMSE, lowering it to 5080 MW. The mean value of peaks across the entire sample is 65 707 MW, so Model 0 has roughly 8% error. Allowing weather covariates in Model 1 lowers the RMSE to 4884, but doing so hourly in Model 2 lowers it further to 2854 MW. Finally, adjusting for serial correlation lowers the RMSE to 2515 MW and the additional Jensen adjustment lowers the RMSE even further to 2417 MW, or only 3.7% error.

5. Conclusion

In order to estimate daily peak electricity demand for the Republic of Korea over 2010–2019, we employ the concept of cross-temperature

response functions first introduced by Chang et al. (2016b). However, instead of economic covariates considered by those and other authors, our cross-TRFs are based on weather-related covariates and allow for variation by time of day. We also develop an adjustment for the Jensen gap introduced by the convexity of the maximum function used to estimate the daily peak demand, which is novel to the best of our knowledge.

Looking at a daily model with pooled hourly data, we find small but significant effects of wind speed, cloud cover, and especially humidity on the relationship between temperature and electricity demand. Without these covariates, demand is underestimated at the hottest temperatures, likely due to a lack of humidity at these temperatures, and overestimated at the coldest temperatures, likely due to a lack of cloud cover at these temperatures. It is precisely at these extreme temperatures when electricity demand peaks.

Table 5

Peak Demand and Deficits of Estimates, Last Week of 2019. Demand deficits in MW over the last week of 2019 for no model, for Models 0, 1, and 2 with no adjustment for Jensen's inequality, Model 2 with a serial correlation adjustment (labeled Model 2+), and Model 2 with both serial correlation and Jensen adjustment (labeled Model 2++).

	Peak Demand	Deficit (+)/Surplus (–)		Model 1	Model 2	Model 2+	Model 2++
		No model	Model 0				
12/25/19 (W)	65 610	5 404	554	2 043	–3540	–3 810	–4 489
12/26/19 (T)	79 431	12 522	8 017	6 575	1674	1 987	1 229
12/27/19 (F)	78 990	12 081	6 528	6 333	998	–1 109	–1 892
12/28/19 (S)	65 014	3 740	–1 916	–1 742	–5761	–5 725	–6 416
12/29/19 (S)	62 540	6 021	1 651	1 177	–2447	–171	–785
12/30/19 (M)	75 633	17 298	13 159	10 044	9727	10 689	10 053
12/31/19 (T)	76 622	9 713	2 178	–2 068	–4816	–9 477	–10 319

The impacts and biases from omitting covariates also vary by time of day. The time-invariant TRF model with no covariates generally overestimates demand response in the morning but underestimates demand response in the afternoon, compared to a model that includes the covariates and allows the demand response to vary over the course of a day even while holding temperature fixed.

Perhaps more importantly, we find that accounting for weather-related covariates improves the explanation of the daily peak demand. Allowing for serial correlation and using a moment estimator of the Jensen gap provides even more accurate numerical explanations of the daily peak over the course of the decade we examine.

CRedit authorship contribution statement

J. Isaac Miller: Conception of research, coding, statistical estimation, most writing, response to referees, proofreading. **Kyungsik Nam:** Conception of research, data management, some writing, response to referees, proofreading.

Appendix A. Adjustments to the data

A.1. Adjustments for missing weather data

We encounter a small number of missing observations in the weather data, summarized in Table A.1. There are no missing observations in the daily data from precipitation and cloud cover, nor are there any missing observations in the electricity demand data. However, there are eleven hours spread across three days with missing temperature, humidity, and/or wind speed data. There is nothing in our data or our knowledge to suggest that September 30, 2013 and June 5, 2018 were unusual in any way, so we can consider them to be missing-at-random.

February 22, 2015 was a Sunday and within three days of Seollal. One can easily imagine a neglect of data collection early the Sunday morning following a major holiday, and, because subsequent analysis will show that electricity demand is reduced on majors holidays, we cannot consider these data missing-at-random.

Because the percentage of missing data in the sample is minuscule and even the percentage of missing data on February 22, 2015 in the sample of days following a major holiday is small, we simply omit these three days from any subsequent analysis. Omitting the entire three days drops 72 h instead of only the 11 missing hours, but given that our sample size without these days is 87,576 h, the omission should have no noticeable impact on our analysis.

If those three days were of particular interest to a researcher, we suggest interpolating the temperature, humidity, and wind speed in a reasonable way, but then using our models estimated without those three days for further analysis. Note that the peak demand on those days occurred at 17:00 on September 30, 2013, 24:00 on February 22, 2015, and 17:00 on June 5, 2018, hours for which weather data are not missing.

Table A.1

Missing Weather Data. X denotes hours missing from temperature (temp), humidity (humd), and wind speed (wind) data.

Date	Time	temp	humd	wind
September 30, 2013	10:00	X	X	
September 30, 2013	11:00	X	X	X
September 30, 2013	12:00		X	
September 30, 2013	13:00	X	X	X
September 30, 2013	14:00		X	X
September 30, 2013	15:00		X	
February 22, 2015	05:00		X	
February 22, 2015	06:00		X	
February 22, 2015	07:00		X	
February 22, 2015	08:00		X	
June 5, 2018	22:00			X

A.2. Adjustments for income

Removing long-run trends in demand that are unrelated to climate change is critical to estimate cross-temperature response functions. Chang et al. (2014, 2016b) stochastically detrend (sectoral) electricity demand simply by measuring log demand relative to a 12-month moving average. Our goal is explicitly on the short-run response of demand, so we similarly want to remove any long-run movement. However, we allow for the possibility that such long-run movement may be driven in part by climate change over the past decade.

For this reason, we do not simply subtract out a moving average, but rather we condition log demand on (annual) log real GDP, obtained from the World Bank.⁵ Because the time period is much less than that of Chang et al. (2014, 2016b), which spans the decades of South Korea's rapid economic development, we do not model the coefficient of the conditioning regression as time-varying. Table A.2 shows the results. The final column shows not the conditional demand, but rather the predicted demand for each year. In estimating peak demand, these must be factored back into the subsequent model output.

By not also detrending the weather variables, we are implicitly assuming no substantive adaptation to climate change over the decade of our sample. In other words, a temperature increase has the same effect over the span of a day as over the span of a decade. This assumption seems more plausible than the reverse: detrending the temperature series would assume perfect adaptation, so that changes in energy efficiency are synced to climate change.

As a reference, Fig. A.1 shows the annually aggregated series, standardized for graphical comparison by subtracting the sample mean and dividing by the sample standard deviation. Importantly, note that there is no obvious trend in what we label excess electricity demand — i.e., with the stochastic trend removed. There is general upward movement in temperature and humidity and general downward movement in wind speed, cloud cover, and precipitation. However, a possible cycle

⁵ Downloaded from <https://data.worldbank.org/indicator/NY.GDP.MKTP.KN?locations=KR> on September 6, 2020.

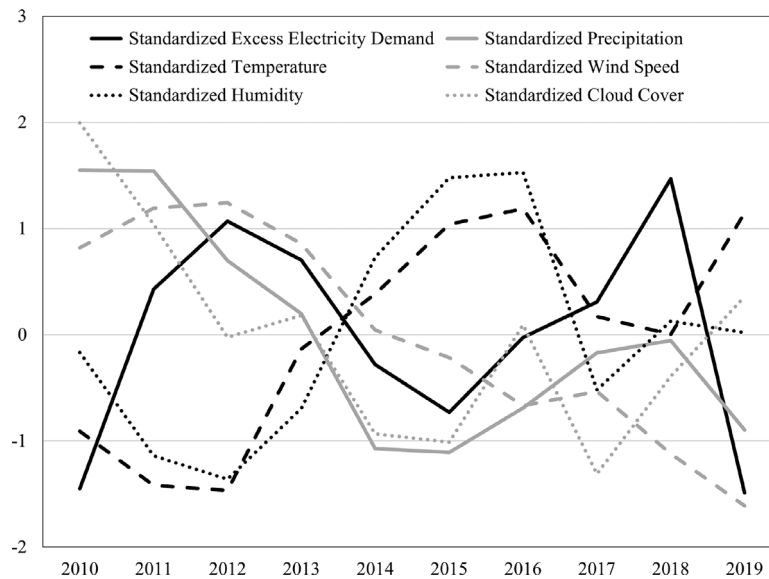


Fig. A.1. Annually Aggregated Series. Annual average of national excess (detrended) electricity demand and annual average of five-city average temperature, humidity, wind speed, cloud cover, and precipitation. Annual series are standardized to have zero mean and unit standard deviation for the purpose of illustration.

Table A.2

Adjustments for Income. Generic regression coefficients β_0 and β_1 are estimated from a regression of log electricity demand onto log real GDP. Excess log electricity demand, used in subsequent analysis, is the series of residuals from this regression. The exponent of the prediction from this regression, given by the last column, is the prediction of an annual average that must be factored back into the final analysis to estimate peak demand at any given time in the respective year.

	Real GDP (trillion KRW)	ln(Real GDP)	Predicted ln(Demand) $\hat{\beta}_1$ $\hat{\beta}_0$	Predicted demand (MW)
			0.642 6.235	
2010	1426.62	7.263	10.897	53 992
2011	1479.20	7.299	10.920	55 261
2012	1514.74	7.323	10.935	56 109
2013	1562.67	7.354	10.955	57 242
2014	1612.72	7.386	10.975	58 412
2015	1658.02	7.413	10.993	59 460
2016	1706.88	7.442	11.012	60 578
2017	1760.81	7.474	11.032	61 800
2018	1807.74	7.500	11.049	62 852
2019	1844.49	7.520	11.061	63 669

appears more clearly than a possible trend — especially in temperature and humidity.

A plausible reason for the evident cycle is the El Niño Southern Oscillation. El Niño events are characterized by temperature peaks in this oscillation. Not only does the 2015–16 period correspond to an El Niño event, but it also corresponds to a super El Niño event. The impact of El Niño on humidity near the Korean peninsula is not well-known, but climate models tend to project more rainfall to the southwest, along the coast of China. It seems plausible that such precipitation would have caused the evident increase in humidity over this time period, though precipitation in Korea itself was lower.

Table A.3 shows the correlation coefficients between the annually aggregated series, which provide a bit more information. Temperature, humidity, and cloud cover are countercyclical with demand, but wind speed and precipitation are procyclical. To a rough approximation, these correlations suggest that annual energy demand in the Republic of Korea is driven more by cold, windy, and clear winter days than it is by sweltering, humid summer days.

All of the other weather series appear to be highly correlated with temperature, with cloud cover having the weakest correlation of nearly 0.5 in absolute value. Of particular note, precipitation has correlations with the remaining weather variables that exceed 0.7 in

Table A.3

Correlations in the Annually Aggregated Data. Sample correlations in the annualized series of excess electricity demand (demd), temperature (temp), humidity (humd), wind speed (wind), cloud cover (clou), and precipitation (prec). Excess electricity demand is calculated as an arithmetic mean of the exponent of excess log electricity demand detrended using log real GDP.

	demd	temp	humd	wind	clou	prec
demd	1.00					
temp	−0.39	1.00				
humd	−0.39	0.83	1.00			
wind	0.22	−0.81	−0.55	1.00		
clou	−0.32	−0.49	−0.34	0.41	1.00	
prec	0.18	−0.90	−0.72	0.71	0.74	1.00

absolute value. These high correlations suggest that the information gained from including precipitation may be largely redundant when the other covariates are included. Because precipitation also has the lowest correlation with demand, it is a candidate for exclusion from the analysis. Indeed, as we explain in the paper, we do exclude it.

A.3. Adjustments for weekends and holidays

Another adjustment that Chang et al. (2016b) make to demand is to account for irregular days along the lines of earlier work by Moral-Carcedo and Vicéns-Otero (2005) and Chang et al. (2014). Such an adjustment is critical in the analysis of sectoral demand, because a major holiday such as Seollal or even just an ordinary Sunday will shift demand away from the industrial and commercial sectors and into the residential sector. Those authors faced the additional complication of relying on billing cycle data, so that such calendar effects had to be accounted for in order to compare calendar months containing billing cycle months ending on different days in the calendar month.

The present analysis avoids most of the complication of these irregular days for several reasons. First, because the data are much more recent, automated meter reading data avoid billing cycle complications. Second, because the emphasis of the present analysis is on aggregate rather than sectoral demand, there is no need to model temporary sectoral shifts. However, these days certainly *can* make a difference in the shape of the temperature response function, for the simple reason that the sectoral TRF need not be the same: industrial demand is insensitive to temperature in comparison with residential and commercial demand (Chang et al., 2016b; Kim et al., 2016).

Table A.4

Adjustments for Weekends and Holidays. Means calculated from regressing excess log electricity demand on binary variable for weekends and holidays described in the text. Subscripts on generic regression coefficients β denote the following: 0: intercept; Hol: holiday; Sum: summer vacation; Sat: Saturday; Sun: Sunday; Mon: Monday; M-2: two days prior to a major holiday; M-1: day before a major holiday; M: major holiday (Seollal or Cheosuk); M+1: day after a major holiday; M+2: two days after a major holiday; M+3: three days after a major holiday.

Coeff: β	Estimate: $\hat{\beta}$	$\exp(\hat{\beta})$	Coeff: β	Estimate: $\hat{\beta}$	$\exp(\hat{\beta})$
β_0	0.050	1.051	β_{M-2}	-0.059	0.943
β_{Hol}	-0.106	0.900	β_{M-1}	-0.252	0.777
β_{Sum}	-0.010	0.990	β_M	-0.352	0.703
β_{Sat}	-0.088	0.916	β_{M+1}	-0.309	0.734
β_{Sun}	-0.169	0.845	β_{M+2}	-0.207	0.813
β_{Mon}	-0.032	0.969	β_{M+3}	-0.084	0.920

Unfortunately, with the exception of weekends, the number of such days in the sample is too small to reliably identify such differences. In preliminary analyses using the TRF only model (Model 0), we experimented with several specifications to account for irregular days, including dropping them outright, but we found little difference in the results. In our final analysis, we simply condition detrended log energy demand on these days and retain the residuals as excess electricity demand for further analysis.

The irregular days we employ were provided by KEEI and are quite similar to those of Chang et al. (2014), also for the Republic of Korea. The main differences are Hangul Proclamation Day, which has become a national holiday only recently, and the so-called sandwich days between weekends and major holidays. Formally, the holidays are New Year's Day, Independence Movement Day, Labor Day, Children's Day, Memorial Day, National Liberation Day, National Foundation Day, Christmas, Hangul Proclamation Day, Buddha's Birthday, election days and the sandwich days and other one-off holidays. Each of these are given equal weight – i.e., a single binary variable for all. A separate binary variable is used for Summer Vacation. Six different binary variables are used for major holidays, Seollal and Chuseok: two days before, one day before, day of, day after, two days after, and three days after.

We include separate binary variables for Saturday, Sunday, and Monday, following Chang et al. (2014). The inclusion of Monday in their analysis was attributed to early Monday morning industrial and commercial demand patterns that differ from the rest of the week. In all, there are eleven binary variables, and Table A.4 shows the results from regressing the log of electricity demand on these binary variables. We note that all coefficient estimates are statistically significant at the 1% level, suggesting that all of these days statistically significantly lower national electricity demand.

Appendix B. Model specification

The parameter vector α_h^k has $p_{hk} + 2q_{hk} + 1$ parameters, meaning that Model 2 in Eq. (8) has $\sum_{h=1}^{24} (p_{hk} + 2q_{hk} + 1) + \sum_{k=1}^K (p_{hk} + 2q_{hk} + 1)$ parameters for all $h = 1, \dots, 24$ with the restriction that $\alpha_1^k = \dots = \alpha_{24}^k$ for $k > 0$ discussed in the paper. Supposing that $p_{hk} = 2$ and $q_{hk} = 1$ for all h and all k , the model would have $5(24 + K)$ parameters, which would be 140 with our four covariates, humidity, wind speed, cloud cover, and precipitation. Given the very large size of our sample, $T = 87,576$ observations, 140 is not unacceptably large. However, there are two problems created by so many parameters.

One problem is the empirical issue of near multicollinearity that often arises with the FFF form. Coefficients tend to explode in magnitude and uncertainty as the marginal contributions of the additional regressors collapse. Although the theory of the FFF form suggests an improvement in the approximation $\tilde{g}_h^k(s)$ as p_{hk} and q_{hk} increase, practice suggests little advantage in setting these numbers very high.

A second problem is the sheer number of permutations of p_{hk} and q_{hk} across h and k . If we use an information criterion to select these,

with just a binary choice between just 0 or 1, we must estimate $2 \times 2 = 4$ models to select $p_{\bullet 0}$ and $q_{\bullet 0}$ for a single time-invariant TRF and $4^5 = 1024$ models to select these orders for both time-invariant TRF and cross-TRFs in the model in Eq. (9). Raising the binary choice to up to five choices, 0, 1, 2, 3, 4, means estimating $25^5 = 9,765,625$ models. And estimating a separate model for each hour of the day as in Model 2 would entail estimating even more models.

We restrict $p_{hk} = 2$ and $q_{hk} = 1$ for $k > 0$ – i.e., for all covariates, thus avoiding the exponentiation in the preceding paragraph. This choice, quadratic plus a pair of trigonometric functions, echoes the finding of Chang et al. (2016b) for their cross-TRFs. As shown in the figures in the paper, it provides quite a flexible specification.

Appendix C. Interval estimates

We do not present t -statistics for any individual coefficients, because their interpretation may be misleading. Semiparametric models using the FFF form may be very nearly perfectly multicollinear even after the optimal order of the approximation is chosen. Near perfect multicollinearity may lead to large and imprecisely measured parameter estimates, as suggested in Table 2 in the paper. However, their linear combination – i.e., the TRF and cross-TRFs – may be precisely estimated. In order to assess uncertainty about TRF and cross-TRFs, interval estimates are constructed using a sieve bootstrap procedure.

For the pooled daily models, Model 1 or Model 0, the bootstrap postulates that the error term in Eq. (9) or (10) is

$$\eta_{t+h/24} = \rho_1 \eta_{t+(h-1)/24} + \dots + \rho_d \eta_{t+(h-d)/24} + e_{t+h/24}$$

with white noise ($e_{t+h/24}$). The value $d = 48$ is chosen so that white noise is both plausible and supported by statistical tests: 24 hours across 2 days of serial correlation. From the fitted series ($\hat{\eta}_{t+h/24}$), we estimate the ρ 's by least squares and obtain a new fitted series ($\hat{e}_{t+h/24}$).

The bootstrap draws a series ($e_{t+2/4}^*$) from these fitted residuals once they are demeaned. Then

$$\eta_{t+h/24}^* = \hat{\rho}_1 \eta_{t+(h-1)/24}^* + \dots + \hat{\rho}_d \eta_{t+(h-d)/24}^* + e_{t+h/24}^*$$

uses the least squares estimates of the ρ 's from the previous step, and

$$y_{t+h/24}^* = x'_{h,t+h/24} \hat{\alpha}_h + \eta_{t+h/24}^*$$

uses the least squares estimates of α_h . The intervals are constructed as the 0.025 and 0.975 quantiles from the distributions of the TRF and cross-TRFs estimated by $\hat{y}_{t+h/24} = x'_{h,t+h/24} \hat{\alpha}_h$ and 999 bootstrap values of $y_{t+h/24}^*$.

The procedure is the same for the hourly models using daily data, but the lags are daily rather than hourly, with the error term of Eq. (8) given by

$$\eta_{h,t} = \rho_1 \eta_{h,t-1} + \dots + \rho_d \eta_{h,t-d} + e_{h,t}$$

with white noise ($e_{h,t}$). We find that $d = 2$ days is not quite enough to eliminate serial correlation for all hours h , but $d = 3$ is adequate.

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